

Closed-Loop Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Energy Absorbers

ASME IDETC-CIE 2026 — Presentation-Only Abstract

Track: DAC-10 — Design of Engineering Materials and Structures

Tensegrity-inspired architectures, in which rigid struts are suspended within a continuous flexible network, can exhibit tunable nonlinear ~~mechanical responses and favorable~~ responses and high strength-to-weight ~~characteristics~~ ratios, motivating their use in lightweight ~~protective and~~ energy-absorbing structures. Multi-material fused deposition modeling (FDM) can ~~now~~ co-print rigid ~~(PLA)~~ PLA struts and flexible ~~(TPU)~~ TPU elements in a single build, but the resulting design space—spanning strut geometry, tension-element cross-section, ~~connectivity~~ topology, and unit-cell tiling—is too large ~~for efficient trial-and-error experimentation~~ to explore by trial and error.

This work presents a closed-loop experimental campaign ~~guided by~~ using Bayesian optimization (BO) ~~for developing to develop~~ multi-material 3D-printed tensegrity-inspired structures ~~optimized for improved~~ energy absorption under quasi-static compression and drop-weight impact ~~loading~~. ~~The key contribution is a design-print-test loop that operates directly on physical measurements without relying on finite-element simulation: a~~ A Gaussian-process surrogate ~~model is trained on experimental measurements of~~ trained on peak transmitted force, specific energy absorption (SEA), and compaction efficiency ~~, and an acquisition function recommends the~~ recommends each next batch of designs ~~after each test round, enabling sequential design selection without finite-element simulation~~. The workflow ~~is largely automated—design generation, surrogate updating, and acquisition-function optimization proceed without operator input, while specimen loading and fixture placement remain the primary manual steps~~ combines automated surrogate updating and experiment selection with manual specimen handling and test setup.

We ~~define a parameterized~~ parameterize a family of unit cells ~~using a core-wrapping strategy~~ in which rigid PLA struts are enclosed by continuous TPU skins to improve interfacial interlocking. Design variables include strut diameter ~~and~~ , length, tension-element width and thickness, ~~number of struts per unit~~ strut count per cell, and tiling pattern. Specimens are fabricated on a ~~consumer-grade~~ multi-material FDM system and tested under quasi-static compression and instrumented drop-weight impact.

We will report ~~the progression of how~~ BO-selected designs evolve across iterations, ~~the evolution of the surrogate model how surrogate predictions change~~ as data accumulate, and ~~what~~ Pareto-efficient trade-offs emerge between SEA and peak ~~transmitted force within the explored design space~~ force. We will also discuss practical lessons for ~~operating a the~~ closed-loop ~~physical experiment~~ workflow, including print-failure handling, ~~batch-to-batch~~ batch variability, and exploration-exploitation balance.